

sequences.

Options :

6406533959741. ✖ 10

6406533959742. ✖ -11

6406533959743. ✔ -3

6406533959744. ✖ 2

Question Number : 41 Question Id : 6406531175143 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Given the k-mers : 'GCCGT', 'CGTAG', 'TAGGC', 'GTAGG', 'CTATG', 'CCGTA', 'TATGC', 'TGCCG', 'AGGCG', 'ATGCC', reconstruct the sequence from which the k-mers were generated.

Options :

6406533959745. ✖ CTATACCGTAGGCG

6406533959746. ✔ CTATGCCGTAGGCG

6406533959747. ✖ ATCGGTACAACGG

6406533959748. ✖ TATGCCGTAGGCGC

Sw Engg

Section Id :	64065382760
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	23
Number of Questions to be attempted :	23
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653181951
Question Shuffling Allowed :	No

Question Number : 42 Question Id : 6406531175148 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE ENGINEERING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533959765. ✓ YES

6406533959766. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653181952

Question Shuffling Allowed :

Yes

Question Number : 43 Question Id : 6406531175149 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following is **not** an Integrated Development Environment (IDE)?

Options :

6406533959767. ✗ Visual Studio Code

6406533959768. ✗ PyCharm

6406533959769. ✗ Thonny

6406533959770. ✓ TextEdit (Mac)

Question Number : 44 Question Id : 6406531175150 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following statements below.

- Statement 1: Executing `git pull` in Git fetches changes from the remote repository and automatically merges them into the current branch.
- Statement 2: Executing `git fetch` in Git retrieves changes from the remote repository and automatically merges them into the current branch.

Identify the option that is correct about the above statements.

Options :

6406533959771. ✓ Statement 1 is correct, Statement 2 is incorrect.

6406533959772. ✖ Statement 1 is incorrect, Statement 2 is correct.

6406533959773. ✖ Both statements are correct.

6406533959774. ✖ Both statements are incorrect.

Question Number : 45 Question Id : 6406531175152 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software engineer is working on fixing an issue with a third-party payment processing service in an e-commerce platform. To identify the problem, the engineer starts by reviewing the API documentation, example codes, etc..

Which of the following debugging strategies is being used by the engineer?

Options :

6406533959779. ✖ Input manipulation

6406533959780. ✖ Backwards strategy

6406533959781. ✖ Forwards strategy

6406533959782. ✔ Blackbox debugging

Question Number : 46 Question Id : 6406531175153 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A hiring manager is conducting an interview with a candidate. Instead of following a predefined set of questions, the manager asks open-ended questions and adapts the follow-up questions based on the candidate's responses, allowing for a free-flowing conversation.

Which type of interview is the manager conducting?

Options :

6406533959783. ✖ Structured interview

6406533959784. ✖ Semistructured interview

6406533959785. ✔ Unstructured interview

6406533959786. ✖ Questionnaires

Question Number : 47 Question Id : 6406531175154 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software development team is building a module for an online store's inventory management system. The module has different inventory-related functions, such as `UpdateStock()`, `GenerateInventoryReport()`, and `SendLowStockNotification()`. All three functions are part of the same module, but they are loosely related in terms of their purpose. The module is performing several distinct tasks: updating the stock, generating reports, and sending notifications. However, the functions in the module are executed one after the other.

Which type of cohesion the module has?

Options :

- 6406533959787. ✖ Coincidental cohesion
- 6406533959788. ✔ Procedural cohesion
- 6406533959789. ✖ Communicational cohesion
- 6406533959790. ✖ Sequential Cohesion

Question Number : 48 Question Id : 6406531175155 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In the design of an e-commerce platform, different modules handle tasks like managing product listings, processing customer orders, and handling payment transactions. While each module focuses on its own task, they all need to access a shared order database to retrieve and update order statuses, customer information, and payment details.

Identify the correct statement about the modules of the given application.

Options :

- 6406533959791. ✔ The modules have high cohesion and are highly coupled.
- 6406533959792. ✖ The modules have low cohesion and are highly coupled.
- 6406533959793. ✖ The modules have low cohesion and are loosely coupled.
- 6406533959794. ✖ The modules have high cohesion and are loosely coupled.

Question Number : 49 Question Id : 6406531175156 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following diagrams models the sequence of message exchanges among parts of a system?

Options :

- 6406533959795. ✖ Class diagram
- 6406533959796. ✖ State machine diagram
- 6406533959797. ✖ Activity diagram
- 6406533959798. ✔ Interaction diagram

Question Number : 50 Question Id : 6406531175157 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the main objective of *sprint retrospective* in agile project management?

Options :

6406533959799. ✖ To decide which work is to be done in the current sprint and how the chosen work will be done.

6406533959800. ✖ To prioritize the work for the development team.

6406533959801. ✖ Team demonstrates what they have completed in the given sprint.

6406533959802. ✔ To evaluate the last sprint - team dynamics, processes, tools etc.

Question Number : 51 Question Id : 6406531175161 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose the risk assessed for budget slippage is 1.8, and the probability of budget slippage occurring during project development is 0.9. What is the impact of budget slippage on project development?

Options :

6406533959815. ✖ 16.6

6406533959816. ✔ 2

6406533959817. ✖ 3

6406533959818. ✖ 0.17

Question Number : 52 Question Id : 6406531175162 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a file-sharing application that allows users to upload and share files with others directly. The files are stored on the users' devices and exchanged directly between users, without relying on a central server for storage or transmission.

What type of software architecture is most likely used for the file-sharing application?

Options :

6406533959819. ✖ Client-server system

6406533959820. ✖ Pipe and filter

6406533959821. ✖ Model view controller

6406533959822. ✔ Peer-to-peer architecture

Question Number : 53 Question Id : 6406531175163 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A company is building a tool for its own developers to streamline the software development lifecycle. The tool will primarily be used for managing version control, tracking bugs, and

automating deployment pipelines.

In this case, which type of clients are we dealing with?

Options :

6406533959823. ✖ External users

6406533959824. ✔ Internal users

6406533959825. ✖ Software or software components

6406533959826. ✖ None of these

Question Number : 54 Question Id : 6406531175164 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a software organization X has developed a complete software product and wants to ensure it works well in real-world scenarios by getting feedback from actual users before the final release.

Identify the type of testing X wants to perform.

Options :

6406533959827. ✖ System testing

6406533959828. ✖ Integration testing

6406533959829. ✖ Alpha testing

6406533959830. ✔ Beta testing

Question Number : 55 Question Id : 6406531175165 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A food delivery app is shared with a predefined set of customers. The customers report an issue where the delivery tracking feature displays incorrect locations for delivery drivers, causing confusion about the estimated delivery time.

This activity is part of which phase of the software development lifecycle?

Options :

6406533959831. ✖ Requirement gathering

6406533959832. ✖ Development phase

6406533959833. ✖ Design phase

6406533959834. ✔ Testing phase

6406533959835. ✖ Maintenance phase

Question Number : 56 Question Id : 6406531175167 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In an agile project, the team is working on implementing a user story. If the user story cannot be completed within one iteration (1-2 weeks), the team decides to subdivide it into smaller, more manageable ones.

What does this approach make the user story?

Options :

- 6406533959840. ✖ Specific
- 6406533959841. ✖ Measurable
- 6406533959842. ✔ Achievable
- 6406533959843. ✖ Relevant
- 6406533959844. ✖ Timeboxed

Question Number : 57 Question Id : 6406531175168 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Imagine a food delivery application where users can search for restaurants, place orders, and track deliveries. However, the application fails to provide any real-time updates about the progress of the order or the delivery.

Which design heuristic is lacking in this scenario?

Options :

- 6406533959845. ✖ Freedom
- 6406533959846. ✖ Flexibility
- 6406533959847. ✖ Recognition over recall
- 6406533959848. ✔ Show status

Sub-Section Number :

3

Sub-Section Id :

640653181953

Question Shuffling Allowed :

Yes

Question Number : 58 Question Id : 6406531175151 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider an e-commerce platform that integrates with multiple payment gateways. The platform has a payment processing system that only accepts payments in USD, but the payment gateways provide transaction amounts in different currencies, such as EUR and INR. The class PaymentProcessor can only handle USD amounts, but the payment gateways provide transaction amounts in incompatible currencies.

Which design pattern can be used to solve the problem?

Options :

- 6406533959775. ✖ Factory design pattern
- 6406533959776. ✖ Builder design pattern

6406533959777. ✓ Adapter design pattern

6406533959778. ✗ Facade design pattern

Question Number : 59 Question Id : 6406531175159 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider a software system designed to manage different types of devices in a smart home. There is an interface `SmartDevice` with methods `turn_on()`, `turn_off()`, `connect_to_network()`, and `play_music()`. This interface is implemented by multiple classes, including `SmartLight`, `SmartThermostat`, and `SmartSpeaker`.

However, the `SmartLight` class implements the `play_music()` method, even though lights have no functionality related to music. Similarly, the `SmartThermostat` class implements the `play_music()` method, despite its primary function being temperature control.

Which SOLID principle is violated by this design?

Options :

6406533959807. ✗ Open-close principle

6406533959808. ✗ Liskov substitution principle

6406533959809. ✓ Interface segregation principle

6406533959810. ✗ Dependency inversion principle

Question Number : 60 Question Id : 6406531175160 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following requirements for a mobile app that allows users to book and manage appointments at a healthcare clinic:

- Requirement 1: The mobile app must be able to support at least 500 simultaneous user logins.
- Requirement 2: The mobile app must allow users to view the available doctors and their appointment slots.
- Requirement 3: The mobile app must notify users when an appointment is successfully booked.
- Requirement 4: The mobile app must be able to process payments for appointments within 3 seconds.

Which of the following statements is true?

Options :

6406533959811. ✗ Requirements 1, 2, and 3 are functional requirements, whereas requirement 4 is a non-functional requirement.

6406533959812. ✓ Requirements 2 and 3 are functional requirements, whereas requirements 1 and 4 are non-functional requirements.

6406533959813. ✗ Requirements 1 and 4 are functional requirements, whereas requirements 2 and 3 are non-functional requirements.

6406533959814. ✖ Requirement 2 is a functional requirement, whereas requirements 1, 3, and 4 are non-functional requirements.

Sub-Section Number : 4
Sub-Section Id : 640653181954
Question Shuffling Allowed : Yes

Question Number : 61 Question Id : 6406531175158 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

Consider the Python code below.

```
def classify_numbers(numbers):  
    classification = {"positive": [], "negative": [], "zero": []}  
    for num in numbers:  
        if num > 0:  
            classification["positive"].append(num)  
        else:  
            if num < 0:  
                classification["negative"].append(num)  
            else:  
                classification["zero"].append(num)  
    return classification
```

What is the cyclomatic complexity of the function `classify_numbers`?

Options :

6406533959803. ✖ 3

6406533959804. ✔ 4

6406533959805. ✖ 5

6406533959806. ✖ 6

Sub-Section Number : 5
Sub-Section Id : 640653181955
Question Shuffling Allowed : Yes

Question Number : 62 Question Id : 6406531175166 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which requirement collection technique involves gathering detailed information from recorded procedures, rules, steps, and regulations pertaining to a specific task or activity?

Options :

6406533959836. ✖ Questionnaires

6406533959837. ✖ Interviews

6406533959838. ✖ Focus group

6406533959839. ✔ Documentations

Sub-Section Number :

6

Sub-Section Id :

640653181956

Question Shuffling Allowed :

No

Question Id : 6406531175169 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (63 to 64)

Question Label : Comprehension

A software company is developing a mobile app for a government agency to monitor and control traffic systems in real-time. The app requires integration with multiple hardware devices, real-time processing, and strict security protocols. The developer team consists of highly skilled professionals.

The values for the constants in the COCOMO model for the different types of projects are as follows:

- Organic project: $a = 2.4$, $b = 1.05$
- Semi-detached project: $a = 3.0$, $b = 1.12$
- Embedded project: $a = 3.6$, $b = 1.20$

Consider the source code to be 1,00,000 lines.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 63 Question Id : 6406531175170 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

What is the initial estimated effort for the project?

Options :

6406533959849. ✖ 303 person-months

6406533959850. ✖ 426 person-months

6406533959851. ✖ 522 person-months

6406533959852. ✔ 905 person-months

Question Number : 64 Question Id : 6406531175171 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Considering an effort adjustment factor = 1.88, what is the final estimated effort for the project?

Options :